A Study in Ind-Coherent Sheaves and Correspondences

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Chapter 1

Motivation

1.1 Ind-coherent sheaves

An important method to study modern representation theory is algebro-geometric. Specifically, For linear algebraic group G, **the category of algebraic representation** $\mathrm{rep}(G)$ **can be interpreted as the category of quasi-coherent sheaves on the classifying stack** BG**.** In this view, many useful functors or operations on G are naturally correspond to the sheaf level. Explicit formulas do not work well when modern representation theory are inevitably involved in higher category.

The most significant mathematic object in this book is Ind-coherent sheaves. Before we define it, we should know the reason to drop the Quasi-coherent sheaves that Lurie used in his higher algebra.

To see the reason, let $\mathfrak{g}$ be the Lie algebra of G with associated formal group $\exp(\mathfrak{g})$. Interpret $\mathfrak{g}$-mod as modules over $U(\mathfrak{g})$ and $\mathrm{rep}(G)$ as modules of compactly supported distributions $\mathcal{E}'(G)$. In this sense, The restriction functor

$$\mathrm{rep}(G) \to \mathfrak{g}\text{-mod}$$

can be interpreted as the pullback functor along the map

$$B(\exp(\mathfrak{g})) \to BG$$

Since one can prove that the category $\mathfrak{g}$-mod is canonically equivalent to $\mathrm{QCoh}(B(\exp(\mathfrak{g})))$. Let us now be given a homomorphism $\alpha : \mathfrak{g}' \to \mathfrak{g}$. The functor of restriction $\mathfrak{g}$-mod $\to \mathfrak{g}'$-mod is the pullback functor of

$$f_\alpha : B(\exp(\mathfrak{g}')) \to B(\exp(\mathfrak{g})). \tag{$\bigstar$}$$

However the left adjoint to restriction functor is induction functor which cannot correspond to the direct image along ($\bigstar$). More explicitly speaking, in the sense of representation theory, the direct image functor as the right adjoint of pullback functor of ($\bigstar$) is coinduction functor. In the case of infinite dimension, **the Frobenius Reciprocity Ind $\cong$ Coind fails to hold and thus the geometric realization of induction functor fades.**

The birth of ind-coherent sheaves is aim to deal with the situation that Frobenius Reciprocity fails. For the case of Frobenius Reciprocity holds, for example classical Lie algebra[1], the ind-coherent sheaves are equivalent to quasi-coherent sheaves.

[1]Here, "classical" means finite dimensional

In the sense of ind-coherent sheaves, the restriction functor along the homomorphism α is given by the !-pullback functor

$$(f_\alpha)^! : \mathrm{IndCoh}(B(\exp(\mathfrak{g}))) \to \mathrm{IndCoh}(B(\exp(\mathfrak{g}')))$$

the functor of induction $\mathfrak{g}'$-mod $\to$ $\mathfrak{g}$-mod will be given by the functor of IndCoh direct image

$$(f_\alpha)_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(B(\exp(\mathfrak{g}'))) \to \mathrm{IndCoh}(B(\exp(\mathfrak{g}))).$$

An analytical analogy is QCoh **behaves more like functions on a space, while** IndCoh **behaves more like measures.**

For every scheme of finite type X, its de Rham prestack X_{dR} is a inf-scheme, one can prove that

$$\mathrm{IndCoh}(X_{dR}) \cong \text{D-mod}(X)$$

Then for an arbitrary inf-scheme $\mathcal{X}$, the ind-coherent sheaf can be given by

$$\mathrm{IndCoh}(\mathcal{X}) = \lim_{Z \to \mathcal{X}} \mathrm{IndCoh}(Z).$$

Hence, one can prove that for any proper map $f : \mathcal{X}' \to \mathcal{X}$, the pair of functor $(f_*^{\mathrm{IndCoh}}, f^!)$ is adjoint. There are several examples to illustrate its advantages.

Example 1.1.1. Take $f : X \to X_{\mathrm{dR}}$ be canonical projection, then the adjoint pair identifies with the pair

$$\mathbf{ind}_{\text{D-mod}} : \mathrm{IndCoh}(X) \rightleftarrows \text{D-mod}(X) : \mathbf{ind}_{\text{D-mod}}.$$

Example 1.1.2. Take $f : Y_{\mathrm{dR}} \to X_{\mathrm{dR}}$, then f_*^{IndCoh} and $f^!$ are correspond to the functors

$$g_{*,\text{D-mod}} : \text{D-mod}(Y) \to \text{D-mod}(X) \quad \text{and} \quad g^!_{\text{D-mod}} : \text{D-mod}(X) \to \text{D-mod}(Y)$$

of D-module push-forward and pullback respectively.

Example 1.1.3. Consider Lie algebra $\mathfrak{g}$ acts on a scheme X. In this case, we have a diagram

$$B(\exp(\mathfrak{g})) \xleftarrow{f_1} B_X(\exp(\mathfrak{g})) \xrightarrow{f_2} X_{\mathrm{dR}}.$$

Then the composite functor

$$(f_2)_*^{\mathrm{IndCoh}} \circ (f_1)^! : \mathrm{IndCoh}(B(\exp(\mathfrak{g}))) \to \mathrm{IndCoh}(X_{\mathrm{dR}})$$

identifies with the localization functor $\mathfrak{g}$-mod $\to$ D-mod(X).

Remark 1.1.4. BG manifests as a geometric representable functor

$$BG : \mathrm{dSch}^{\mathrm{op}} \to \infty\text{-Grpd} \quad X \mapsto \mathrm{Bun}_G(X).$$

From an operadic perspective,

$$\mathrm{Bun}_G(X) \simeq \mathrm{Alg}_X(BG) \simeq \mathrm{Map}_{\mathrm{PreStk}}(X, BG)$$

where we interpret X as an unary ∞-operad and $\mathrm{Bun}_G(X)$ as the category of BG-valued algebras over X_{dR}. When we consider the de Rham prestack X_{dR}, we obtain that

$$\mathrm{LocSys}_G(X) \simeq \mathrm{Alg}_{X_{\mathrm{dR}}}(BG) \simeq \mathrm{Map}_{\mathrm{PreStk}}(X_{\mathrm{dR}}, BG).$$

$B_X(G)$ is the ∞-categorical relative tensor product $X \otimes_G *$.

1.2 Construction of ind-coherent sheaves

There is a heuristic formula we have presented

$$\mathrm{IndCoh}(\mathcal{X}) = \lim_{Z \to \mathcal{X}} \mathrm{IndCoh}(Z) \tag{$\bigstar$}$$

In what sense is this limit taken? In triangulated categories, colimits and limits are not well-behaved.

We do it in the ∞-category of presentable k-linear stable ∞-categories, or equivalently, in the ∞-category of differential graded categories: categories enriched over the category of chain complexes of k-modules, such that the differential satisfies the Leibniz rule with respect to composition. We denote it by DGCat.

For a scheme Z, the category $\mathrm{IndCoh}(Z)$ is the renormalization of $\mathrm{QCoh}(Z)$. One can prove that ind-coherent sheaves have the following properties:

a. DG category $\mathrm{IndCoh}(Z)$ contains all infinite coproducts

b. For a map $Z' \xrightarrow{f} Z$, the functor $\mathrm{IndCoh}(Z') \xrightarrow{f^!} \mathrm{IndCoh}(Z)$ preserves all infinite coproducts.

We can see that the shriek pullback functor $f^{!,\mathrm{QCoh}}$ fails to satisfy property (b) even in the case that f is proper. This is because the direct image functor f_* as the left adjoint of $f^{!,\mathrm{QCoh}}$ does not preserve compactness, and thus its right adjoint does not preserve infinite coproducts according to Brown representability. Indeed, f_* does not necessarily send perfect complexes on Z' to perfect complexes on Z unless f is of finite Tor-dimension. An easy example is $f : \mathrm{Spec}(k) \to \mathrm{Spec}(k[t]/t^2)$. The perfect complex k does not has finit resolution as a $k[t]/t^2$-module.

Fortunately, when f is proper, the direct image functor f_* sends the category of bounded coherent complexes $\mathrm{Coh}^b(Z')$ to $\mathrm{Coh}^b(Z)$ and we define ind-coherent sheaves as the ind-completion of $\mathrm{Coh}^b(-)$

$$\mathrm{IndCoh}(-) := \mathrm{Ind}(\mathrm{Coh}^b(-))$$

The compact objects in the category $\mathrm{IndCoh}(Z)$ are exactly all bounded coherent complexes, and therefore f_* is well-behaved.

The difference of QCoh and IndCoh is quite small to some extend. More clearly, the tautological embedding $\mathrm{Coh}(Z) \hookrightarrow \mathrm{QCoh}(Z)$ induces

$$\Psi_Z : \mathrm{IndCoh}(Z) \to \mathrm{QCoh}(Z)$$

and the ususal t-structure on $\mathrm{Coh}(Z)$ induces one on $\mathrm{IndCoh}(Z)$. One can prove that Ψ_Z is t-exact resulting

$$\mathrm{IndCoh}(Z)^{\geq -n} \to \mathrm{QCoh}(Z)^{\geq -n}$$

is an equivalent. Therefore, the difference occure only at $-\infty$. The difference of f_*^{IndCoh} and f_* is also small. Explicitly, f_*^{IndCoh} is the unique factor that perserves infinite direct sums and makes the diagram

$$\begin{tikzcd}
\mathrm{IndCoh}(Z')^{\geq -n} \arrow[r, "\Psi_{Z'}", "\sim"'] \arrow[d, "f_*^{\mathrm{IndCoh}}"'] & \mathrm{QCoh}(Z')^{\geq -n} \arrow[d, "f_*"] \\
\mathrm{IndCoh}(Z)^{\geq -n} \arrow[r, "\Psi_Z", "\sim"'] & \mathrm{QCoh}(Z)^{\geq -n}
\end{tikzcd}$$

commutes for every n.

The main object of this book is the functor

$$\mathrm{IndCoh}^!_{\mathrm{Sch}} : (\mathrm{Sch})^{\mathrm{op}} \to \mathrm{DGCat}. \tag{$\mathbf{\Pi}$}$$

The construction of it as follows. For a morphism $f : Z' \to Z$, we can factor it as

$$Z' \xrightarrow{f_1} \overline{Z'} \xrightarrow{f_2} Z$$

where f_1 is an open embedding and f_2 is proper. Then we can define

$$f^! := f_1^! \circ f_2^! : \mathrm{IndCoh}(Z) \to \mathrm{IndCoh}(Z')$$

where $f_2^!$ is the right adjoint of $(f_2)_*^{\mathrm{IndCoh}}$ and $f_1^!$ is the left adjoint of $(f_1)_*^{\mathrm{IndCoh}}$.

Since we are discuss in the sense of infinite categories, we need to supply the assignment

$$f \rightsquigarrow f^!$$

with a homotopy-coherent system of compatibilities for n-fold composition of morphism. In a word, we need to show the functoriality in the sense of ∞-categories.

For example, we need to consider the compatibility betweend functor ($\mathbf{\Pi}$) and functor

$$\mathrm{IndCoh} : \mathrm{Sch} \to \mathrm{DGCat} \quad Z \rightsquigarrow \mathrm{IndCoh}(Z) \quad (Z' \xrightarrow{f} Z) \rightsquigarrow (\mathrm{IndCoh}(Z') \xrightarrow{f_*^{\mathrm{IndCoh}}} \mathrm{IndCoh}(Z)).$$

Namely, we need to show whenever we have a Cartesian square

$$\begin{tikzcd} Z_1' \arrow[r, "g'"] \arrow[d, "f_1"'] & Z' \arrow[d, "f"] \\ Z_1 \arrow[r, "g"'] & Z \end{tikzcd}$$

There is a canonical isomorphsim of functors, called base change:

$$(f_1)_*^{\mathrm{IndCoh}} \circ (g')^! \cong g^! \circ f_*^{\mathrm{IndCoh}}.$$

The appearance of the Cartesian square make us inevitably use ∞-categories. For example, consider the diagram

$$\begin{tikzcd} \mathrm{pt} \times_{\mathbb{A}^1} \mathrm{pt} \arrow[r] \arrow[d] & \mathrm{pt} \arrow[d] \\ \mathrm{pt} \arrow[r] & \mathbb{A}^1 \end{tikzcd}$$

the fiber product $\mathrm{pt} \times_{\mathbb{A}^1} \mathrm{pt}$ cannot be taken to be the point-scheme. Rather it makes sense in the derived setting:

$$k \otimes_{k[t]} k = k[\varepsilon] \quad \deg(\varepsilon) = -1.$$

Chapter 2

Derived Algebraic Geometry

Let us revisit concepts from Derived Algebraic Geometry.

1. 1-Cat denote the $(\infty, 1)$-category of $(\infty, 1)$-categories.

2. Spc denote the $(\infty, 1)$-category of spaces. There is a canonical faithful embedding

$$\mathrm{Set} \hookrightarrow \mathrm{Spc}$$

with a left adjoint

$$\mathcal{S} \mapsto \pi_0(\mathcal{S}).$$

3. $\mathrm{Spc} \hookrightarrow \text{1-Cat}$ is a full subcategory with a right adjoint

$$\mathbf{C} \mapsto \mathbf{C}^{\mathrm{Spc}},$$

which discarding non-invertible morphisms.

4. For $\mathbf{c}_0, \mathbf{c}_1$ are objects of $\mathbf{C} \in \text{1-Cat}$. The mapping space denoted by $\mathrm{Maps}_{\mathbf{C}}(\mathbf{c}_0, \mathbf{c}_1) \in \mathrm{Spc}$.

5. Let $\mathbf{C}^{\mathrm{ordn}}$ denotes the homotopy category $\mathrm{Ho}(\mathbf{C})$. Let $\text{1-Cat}_{\mathrm{ordn}}$ form by ordinary categories.

6. a map $\phi : \mathbf{c}_0 \to \mathbf{c}_1$ is said to be an isomorphsim if $\pi_0(\phi)$ is an isomorphsim.

7. Let $\mathrm{Funct}(\mathbf{D}, \mathbf{C})$ denotes $(\infty, 1)$-category of functors. We have

$$\mathrm{Funct}(*, \mathbf{C}) \cong \mathbf{C}$$

and

$$\mathrm{Maps}_{\text{1-Cat}}(\mathbf{D}, \mathbf{C}) = \mathrm{Funct}(\mathbf{D}, \mathbf{C})^{\mathrm{Spc}}.$$

8. A functor $F : \mathbf{C} \to \mathbf{D}$ is said to be an equivalence if it induces an isomorphsim in $(\text{1-Cat})^{\mathrm{ordn}}$.

9. For